Cold Open: finding the 40 seconds that make a stranger stay

TODO(Kartik): name Roll number TODO(Kartik): roll no. Branch TODO(Kartik): branch Kuku AI Generalist, Problem 1

Demo TODO(Kartik): demo link Code TODO(Kartik): repo link

Why this problem

TODO(Kartik) Why this problem: 3–4 sentences in your own words (why promo selection is the bottleneck for a microdrama launch, and what a bad promo costs).

What it does

Cold Open reads a show's screenplay (timestamped episode scripts) and returns a ranked **launch slate**: a hero promo and 4 challengers, each a 20–60 s cut that hooks in the first 3 s, ends on an unresolved beat and makes sense to a stranger. Editors review them in a web cutting room and export CMX3600 EDLs and a CSV cut list.

The rule that shapes everything: **AI judges meaning, code owns facts, the editor makes the final call.** The model only ever names line IDs; it never writes a timestamp, a duration or a rank.

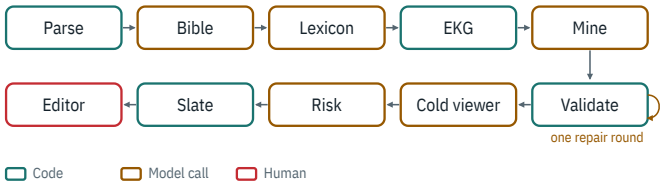
Assumptions

- A promo is 20–60 s (aim 30–45), built from 1–3 segments, with the hook inside the first 3.0 s.
- The paywall sits at episode 10; a promo should land a viewer with free episodes ahead.
- Text screenplays only: no video, music or captions. Cuts snap to shot changes in the script.
- Meta is the primary paid channel; ad-policy risk follows Meta's advertising standards.
- The editor has the final say; every approve, trim and reject is logged.

Who does what

Who	Does	Why them
AI	Show bible and alias resolution, emotion lexicon, picks candidate moments by line ID, one repair attempt, cold-viewer test, grading, ad-policy and spoiler reading	Meaning: who is who, what a stranger gets
Code	Parsing, tension signal, every hard rule (IDs, ≤3 s hook, 20–60 s, no mid-line cuts, verbatim quotes), snapping to shot changes, scoring, diverse selection, exports	Facts must be exact; rules are asserted
Editor	Approves for paid or organic, trims with live re-validation, rejects with a reason	Taste and brand; every call is logged

How it works



Nine stages. Teal is code, amber is a model call, red is the human. The validator sends a failing candidate back once with its errors, then rejects it with a reason code.

- **Bible.** One call turns loglines, speaker labels and code-detected spelling drift into characters (with episode-scoped aliases), arcs, reveals and true continuity conflicts.
- **Tension EKG.** Per-second loudness from emotion tags, punctuation, violent verbs, sound and camera cues: cheap recall, deliberately blind to meaning.
- **Mining.** Per episode the miner proposes 3–6 candidates by line ID, including one quiet line away from the loud peaks; a second pass bridges moments across episodes.
- **Cold Viewer.** A different model sees only what a scroller sees (faces described, never named) and says whether it would stop and what it understood; a grader checks that against the bible.
- **Slate.** A transparent weighted score minus policy, spoiler, paywall and unexplained-name penalties; MMR picks a diverse slate so five promos are not five takes of one slap.

Three choices worth defending

- **IDs, not timestamps.** Checked by ablation: see page 2.
- **A cold viewer from another model family**, so the judge does not share the miner's blind spots or its knowledge of the plot.
- **Replayable runs.** Every call is cached by content hash; a run replays offline and swapping a model is one line in config.yaml.

Results on the full pack

14 episodes, 640 timed lines. The miner proposed **62** candidates; **4** were fixed by one repair round, **0** rejected, **24** removed as near-duplicates, leaving **38** valid promos (100% pass every hard rule, asserted in code). The slate:

#	Cut	Hook	Last line	Engine	Why
1	Flash-forward E11 00:19.40–00:22.06 E10 00:09.00–00:49.74 43 s	`Search her pockets. She is a thief!`	`Order! I will have order in my hall!`	Family betrayal	[mock] Opens on a charged line at a tension peak and cuts before the reply.
2	Straight E9 00:19.00–01:07.74 49 s	`Every wedding needs a widow. Why not two?`	`Every wedding needs a widow. Why not two?`	Forbidden scandal	[mock] Opens on a charged line at a tension peak and cuts before the reply.
3	Straight E3 00:27.00–01:17.60 51 s	`Search her pockets. She is a thief!`	`Lord Ashvale? That is impossible.`	Framed, condemned	[mock] Opens on a charged line at a tension peak and cuts before the reply.
4	Flash-forward E2 00:00.40–00:03.44 E1 00:10.00–00:51.00 44 s	`Every wedding needs a widow. Why not two?`	`Lord Ashvale? That is impossible.`	Humiliation reversal	[mock] Opens on a charged line at a tension peak and cuts before the reply.
5	Flash-forward E8 00:17.40–00:20.44 E7 00:10.00–00:48.74 42 s	`Your father paid me to watch over you.`	`The ring smells of your rose oil, Selene.`	Other	[mock] Opens on a charged line at a tension peak and cuts before the reply.

How I know it's good

TODO	TODO	TODO	100%
hit rate vs gold (run coldopen eval)	precision@10 (coldopen eval --rate)	median cut error vs gold	of final promos pass every hard rule

TODO(Kartik) Run `coldopen ablate` and `coldopen eval --ablate` to fill the ablation line.

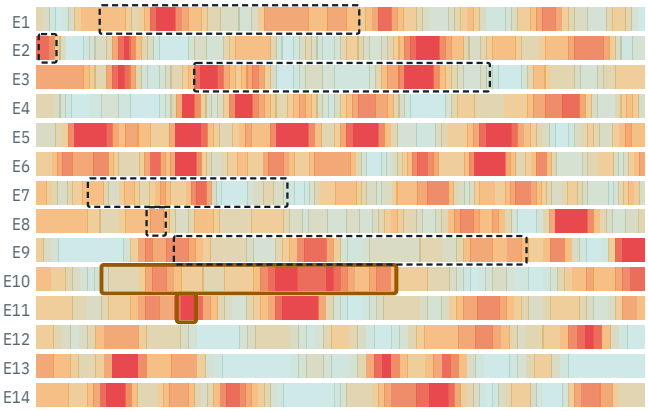
Where it fails

- **Stranger confused** (E3 00:52.00–01:27.74): Cold Viewer comprehension 0.38. [mock] keyword overlap only
- **Stranger confused** (E5 00:06.40–00:09.44; E4 00:24.00–00:58.00): Cold Viewer comprehension 0.38. [mock] keyword overlap only
- **Stranger confused** (E5 00:22.00–01:06.00): Cold Viewer comprehension 0.38. [mock] keyword overlap only

TODO(Kartik) Confirm these three by opening them in the cut room; replace any that are not real failures.

Scale and cost

One show: **152** model calls, **84,093** tokens, **cost unknown (prices.yaml not filled)**, 0 s of wall time at 10 requests per minute. At 150 shows a month that is about 13 M tokens and an unknown cost (fill prices.yaml); the cache makes re-runs after a fix free.



Tension across all 14 episodes (one row each, time normalised per episode). Amber outline: hero; dark dashed: challengers.

How we'd know it worked live

A/B the slate against editor-picked promos on equal spend: 3-second view rate, cost per install, day-1 paid conversion. Ship when editors accept 60%+ of slate promos without re-cutting.

Tools used and where they got it wrong

TODO(Kartik) Name the tools you used, in your own words. Suggested cases from docs/BUILD_LOG.md: Mock provider guessed list semantics; Rate limiter tied to one event loop

Next

Feed editor decisions back to the miner as examples; read the video for real shot changes; learn the score weights from install data instead of defaults.